

16-bit for low power, high performance

While 8-bit microcontrollers dominate the existing embedded market and specific applications, 16-bit microcontrollers strike the cost and performance balance for many handheld and portable devices. Almost all the 16-bit MCUs from different vendors are targeted at portable embedded applications that require decent performance at low power.

Unlike 8-bit and 32-bit, there is no standard core in 16-bit space. Proprietary cores offered by different vendors are targeted at different application areas. Renesas' M16C multi-function MCU family meets the increasing functionality demands corresponding to different market segments. It is compatible with both the low-end as well as high-end MCUs in the family. Electronic household appliances, white goods, audio equipment, TV sets and cameras are some of the application areas.

Mixed-signal processors family MSP430 from Texas Instruments is popular in India. Shailesh Thakurdesai, business development manager for microcontrollers, TI, shares, "As cost-sensitive applications are also demanding more functionality from microcontrollers, the 16-bit RISC architecture of MSP430 provides greater performance, peripherals and low power at no additional cost." He adds, "With ease of programming, modern architecture, in-system emulation capabilities and code compatibility, the migration of code to higher-performance devices is painless." MSP430 finds application in metering units, handheld medical equipment and automation.

16-bit MCUs with digital signal controllers are largely suitable for motor control applications. "System-based chips address the electromagnetic capability robustness for in-vehicle networks, while SmartMOS optimises power and cost for electronic control units in automotive applications," shares Vivek Tyagi, country manager-sales for Freescale Semiconductor.

In spite of the recent launches to keep 16-bit alive, experts feel that 8-bit and 32-bit are the answer for the microcontroller market. "Development is happening to push the performance of 8-bit on the higher end and 32-bit on the lower end. This has squeezed the market share for 16-bit," says Arnaud Julienne, director-MCU & MPU, STMicroelectronics. Although 8-bit might not be growing in terms of revenue, it has shown growth in the number of units sold.

According to market research firm iSuppli, Renesas (now Renesas Electronics) ranks first in sales of 16-bit MCUs. Infineon, Panasonic, Texas Instruments and Freescale are next in the list.

ARM for 32, the choice of many

For applications demanding high levels of performance and integration, the choice is likely to be the 32-bit core based microcontrollers. "Lately, 32-bit MCUs have started dominating the sector just because vendors are able to offer 32-bit performance at competitive price point," shares Patel. Most of the 32-bit MCUs are 32-bit embedded processors connected to peripherals and memory for controller applications. 32-bit MCUs greatly benefit from the tools, software and knowledge base for established PC processors.

In 32-bit space, so far the outright leader is ARM Cortex based microcontroller. Cortex series meets the demand for performance and power. In ARM Cortex-M series, where 'M' stands for microcontrollers, different variants like Cortex-M0, M3 and most recent M4 serve speeds ranging from 50 MHz up to 150 MHz, which is the general MCU requirement.

Cortex-M3 based microcontrollers provide 40-50 per cent savings in code space, very low active power and attractive price point of 65 cents. Going further, Cortex-M4 offers an exciting blend of high-performance MCU with digital signal processing in an easy-to-use single ARM architecture and software ecosystem.

In proprietary cores, Renesas Electronics is growing with its 32-bit product portfolio. "We have observed that there is a growing interest toward higher-MIP 32-bit MCUs, such as our H8SX, R32C and SH2(A). Renesas (now Renesas Electronics) has also created a new 32-bit core named as 'Renesas eXtreme' (RX). This CPU core can achieve 200 MHz, 1.65 DMIPS/MHz and 0.03 mA/MHz," shares Wee Seng.

Atmel's AVR, Freescale's Coldfire and Infineon's Tricore compete with ARM in specific application areas. But market share is the largest for ARM core.

In support of standard cores, Jain shares, "First, there are more than one vendor that can supply the MCU. For example, ARM7-based microcontrollers are available from a number of companies including NXP (LPC series) and Atmel (SAM7). Similarly, 8051 is also available from a number of companies. Second, porting of the code becomes much easy. Nowadays, coding is done in a high-level language (such as 'C'), and the compiler does a lot of job for you. Even with this kind of coding, porting is not easy, as the architecture-specific code (for initialisation of peripherals) needs to be rewritten completely."

"Standard cores generally have most of the interfaces proven as they need to work with some standard interfaces. Hence designing is faster and more dependable. On the other hand, proprietary cores are tailor-made for some specific requirements and the maturing time for them is usually longer. However, these have their own benefits like design security and customisation and will be more optimal for particular designs," says Binu.

In a nutshell, standard cores create a platform that is easy to approach, easy to use and easy to migrate. This could be the reason for the success of 8-bit 8051 and 32-bit ARM. "If you look at 8-bit space, 8051 has achieved its position in standard cores. There is possibly no room for new standard cores in 8-bit space. Slowly the 16-bit slot is migrating to 32-bit space. And in 32-bit, ARM has gained popularity with its core Cortex-M," shares Upendra Patel, chief technology officer at elfnchips.

The favouritism for standard cores makes them score over proprietary ones. But it's the ecosystem around the core that makes it successful. Rather than going through the learning curve to become expert in one, designers can benefit from resources like online community where they can seek help for their designs. A common and consolidated platform seems to be the answer for fast-evolving embedded needs. ■

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